EXPERIMENT No. — 2.

1) **Aim**: To determine the acceptance angle and numerical aperture of an optical fibre using a LED light source (diode laser)

2) **Apparatus Required**: Diode laser, optical fiber, fiber optic chuck, screen, graph paper, measuring tape.

3) **Formula Used**:

Numerical Aperture (NA) = $\sin(\theta_0)$

where, Acceptance angle $\theta_0 = \tan^{-1}\left(\frac{D}{2L}\right)$

4) **Theory**: Numerical aperture is a basic characteristic of specific fiber. It can be thought of as representing the size or "degree of openness" of the input acceptance cone. The light gathering power or flux carrying capacity of a fiber is numerically equal to the sine of the aperture, which is the ratio between area of a unit sphere with the acceptance cone and the area of the hemisphere (2π solid angle). A fiber with a numerical aperture of 0.66 has 43% as much flux carrying capacity as a fiber with a numerical aperture of 1

i.e. $(0.66)^2/(1.0)^2 = 0.43$

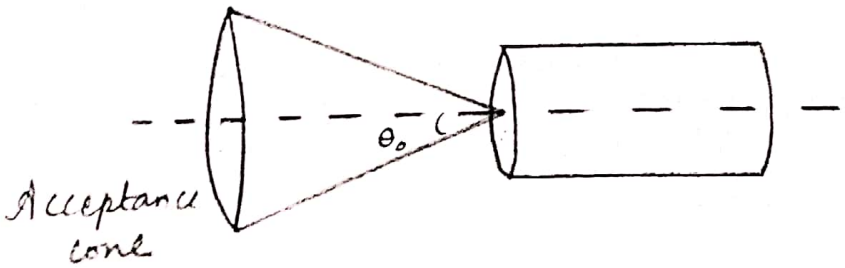
Snell's law can be used to calculate the maximum angle within which

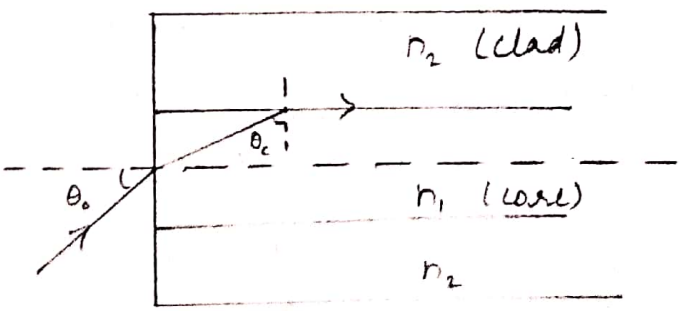
light will be accepted into and conducted through a fiber.

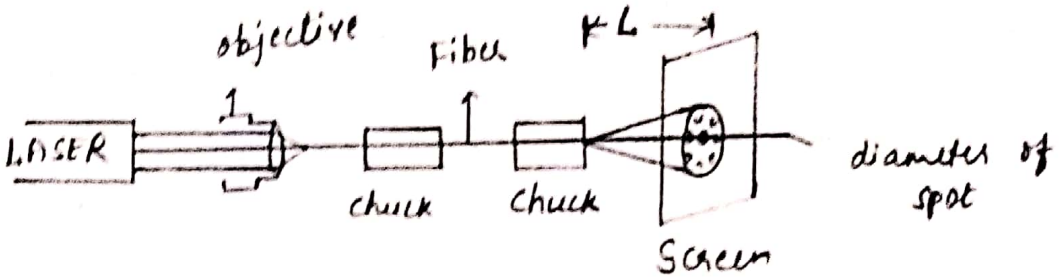
$$NA = \sin\theta_0 = \sqrt{n_1^2 - n_2^2}$$

where, $\sin\theta_0$ is numerical aperture. n_1 and n_2 are the refractive indices of the core and the clad. The semi angle θ_0 of the acceptance cone for a step index fiber is determined by the critical angle θ_c $(= \sin^{-1}(n_1/n_2))$ for total internal reflection to take place at core clad interface.

5) <u>Diagram</u>:







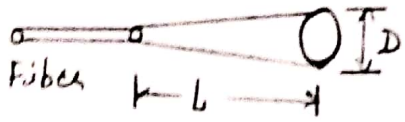


Fig. → Numerical Aperture of Fiber

6) Observation Table:

S.No.	Diameter of spot (D)	Distance between fiber end and screen (L)	$\tan^{-1}\left(\frac{D}{2L}\right)$	$\sin\left(\tan^{-1}\frac{D}{2L}\right)$
1.	1.8 cm	10 cm	5.142	0.0896
2.	2.7 cm	15 cm	5.142	0.0896
3.	3.8 cm	20 cm	5.426	0.0945

4) Calculation:

I Case:- $\tan^{-1}\left(\frac{1.8}{2\times10}\right) = \tan^{-1}\left(\frac{1.8}{20}\right) = 5.142$

$\sin(5.142) = 0.0896 \longrightarrow (i)$

II Case:- $\tan^{-1}\left(\frac{2.7}{2\times15}\right) = \tan^{-1}\left(\frac{2.7}{30}\right) = 5.142$

$\sin(5.142) = 0.0896 \longrightarrow (ii)$

Scanned with OKEN Scanner

III Case: $\tan^{-1}\left(\frac{3.8}{2\times 20}\right) = \tan^{-1}\left(\frac{3.8}{40}\right) = 5.426$

$\sin(5.426) = 0.0945 \rightarrow (iii)$

Averaging eq^n (i) (ii) and (iii) we get

$$\text{Mean NA} = \frac{0.0896 + 0.0896 + 0.0945}{3}$$

$$\boxed{NA = 0.0912}$$

Acceptance Angle $(\theta_0) = \sin^{-1}(NA)$

$= \sin^{-1}(0.0912)$

$= 5.234$

8) Percentage Error :-

Std. Value of NA = 0.1

Cal. Value of NA = 0.0912

$$\text{Percentage Error} = \frac{\text{Std} - \text{Cal.}}{\text{Std.}} \times 100\%$$

$$= \frac{0.1 - 0.0912}{0.1} \times 100\%$$

$= 8.8\%$

9) Result :- The Numerical Aperture of a given fiber is 0.0912 calculated and its acceptance angle is found to be 5.234.

Scanned with OKEN Scanner

10) Precautions:-

(a) Coupling of laser of the fiber should be very proper.

(b) Fiber should be kept straight.

(c) Diameter of spot should be measured accurately.

(d) Direct viewing of laser light should be avoided.

Scanned with OKEN Scanner